Soapomorph Animatronic Project

Course Module: Robotic Modelling and Drone Skin Design.
Module Number: COMP60019.
Lecturer: Luke Emery.

Group Members:
Hannah Arnold
Balraj Bhuller
Chris Mead
Max McGill
Rachel Kirkham

Contents

1.0 Introduction

This document specifies the details of Group A's *Robotic Modelling and Drone Skin Design* project artefact, including its purpose, design and target environment, while considering its effects on stakeholders, the environment, and those involved in its development.

1.1 Idea Overview

The project's artefact is a soap dispenser, designed to look like H. R. Giger's *Xenomorph* design from the *Alien* franchise. The soap is to be dispensed in such a way as to create the illusion of the model's mouth salivating. A tongue featuring an inner jaw will shoot from the mouth in a manner like the creature from the films and dispense soap from the end.

Being a soap dispenser, it is intended to be placed in a bathroom in an entertainment venue, such as the Madame Tussauds wax museum in London.

1.2 Justification.

Entertainment venues, as the name suggests, seek to entertain their guests for the duration of their time there. This endeavour need not be completely paused when nature calls and the bathrooms need not simply be bland boxes with corporate looking equipment. Provided that care is taken to adhere to health and safety concerns, creative flair can be applied to the facilities within the bathrooms, with one of the more straightforward options being an animatronic model that dispenses soap to replace a boring, corporate-looking soap dispenser. Given a well-chosen venue, the *Soapomorph* could bring a memorable experience to guests, especially when that experience is had somewhere unexpected.

2.0 Stakeholder

A stakeholder is a person or group that are affected or have an interest in the actions completed by a business. There are 2 main types of stakeholders, internal and external. Internal stakeholders are directly part of the decision-making part of the business and are often working for it. However, external stakeholders are affected by the business but aren't directly linked to the decision-making process of the business. (BBC Bitesize 2026)

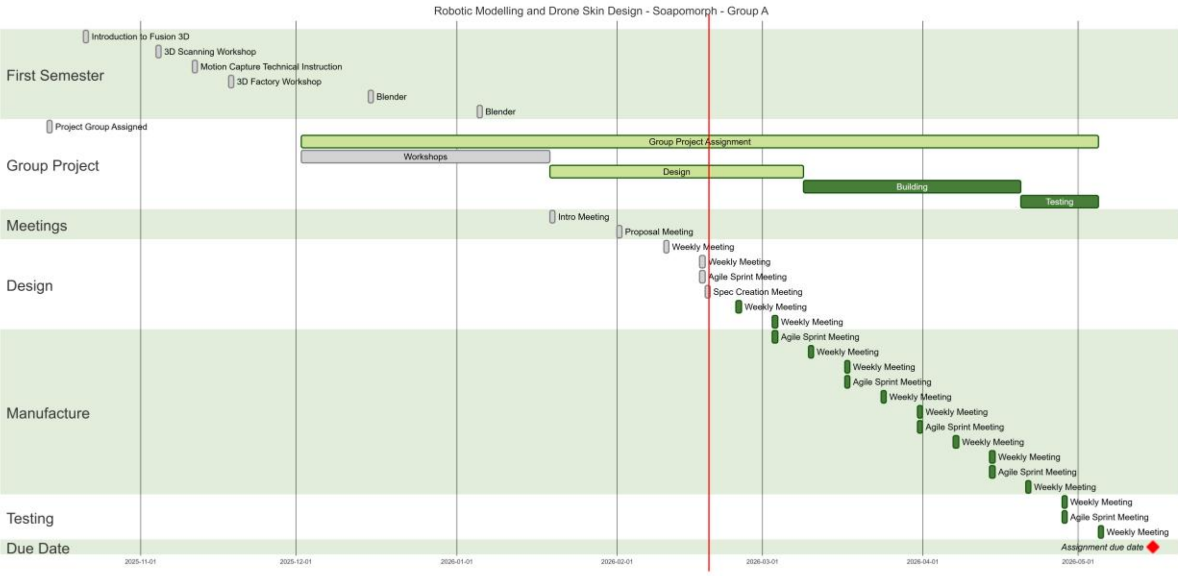
For this project the chosen audience is Madam Tussauds, specifically within the Alian section. This would make the internal stakeholders are:

ID	Name	Position	Company
1	Fiona Eastwood	CEO	Merlin Entertainments
2	Natasha Breeze	Director	Merlin Entertainments
3	Colin North Armstrong	Director	Merlin Entertainments
4	Andrew Car	Director	Merlin Entertainments
5	Fiona Jane Rose	Secretary	Merlin Entertainments
6	Amy Bailey	Secretary	Merlin Entertainments
7	N/A	Managers	Merlin Entertainments
8	N/A	Employees	Merlin Entertainments
9	Luke Emery	Lecturer	University of Staffordshire
10	Hannah Arnold	Student	University of Staffordshire
11	Rachel Kirkham	Student	University of Staffordshire
12	Christopher Mead	Student	University of Staffordshire
13	Max McGill	Student	University of Staffordshire
14	Balraj Bhuller	Student	University of Staffordshire
15	Jahmiel	Student	University of Staffordshire

External Stakeholders:

- Customers
- Suppliers
- Local Community
- Pressure Groups
- Government

3.0 Time plan.



Robotic Modelling and Drone Skin Design - Soapomorph - Group A
First Semester
Introduction to Fusion 3D
3D Scanning Workshop
Motion Capture Technical Instruction
3D Factory Workshop
Blender
Blender
Project Group Assigned
Group Project
Group Project Assignment
Workshops
Design
Building
Testing
Meetings
Intro Meeting
Proposal Meeting
Weekly Meeting
Weekly Meeting
Agile Sprint Meeting
Spec Creation Meeting
Design
Weekly Meeting
Weekly Meeting
Agile Sprint Meeting
Manufacture
Weekly Meeting
Weekly Meeting
Agile Sprint Meeting
Weekly Meeting
Weekly Meeting
Agile Sprint Meeting
Weekly Meeting
Weekly Meeting
Agile Sprint Meeting
Weekly Meeting
Testing
Weekly Meeting
Agile Sprint Meeting
Weekly Meeting
Due Date
Assignment due date
2025-11-01
2025-12-01
2026-01-01
2026-02-01
2026-03-01
2026-04-01
2026-05-01

4.0 Environment.

The Soapomorph is a one-off prototype, meaning its environmental footprint differs from that of a mass-produced commercial product. However, responsible considerations of tis environmental impact remains important across each stage of its life, from the sourcing of materials through to its eventual disposal. As a prototype developed for placement in a public-facing venue such as madame Tussauds, demonstrating environmental awareness in the design and disposal process reflect positively on both the project team and the venue.

4.1 Lifecycle.

The lifecycle of the Soapomorph prototype can be broken down into 3 stages: production, use, and end of life.

During its **production** stage, we use a range of materials across different components. The internal structural frame and mechanical components are manufactured from stainless steel. The production of stainless steel is energy-intensive due to the smelting and alloying of iron, chromium, and nickel. However, since only a small quantity is used for a single unit, the overall impact is limited. The outer casing progressed through several plastic materials depending on the stage of prototyping. PLA is used for early-stage prints due to its ease of printing and biodegradability; PETG and ABS are used for functional testing prototypes due to their improved impact resistance and thermal stability; and ABS or polycarbonate is used for the final casing due to their durability and finish quality. Each of these materials carries different environmental profiles. PLA being the most environmentally friendly as it is plant-derived and compostable under the right condition, while ABS and polycarbonate are petroleum-based and carrya higher carbon footprint. Electrical component including the microcontroller, motor, wiring, and sensors also require energy and raw materials which includes doper and rare earth elements. However, again due to the quantity involved in a single prototype, the actual carbon footprint is small.

During its **Use** phase, the Soapomorph has a minimal environmental impact. The motor and control electronics consume a small amount of electricity, operating only in short bursts when triggered by a user. The environmental impact of the use phase is therefore determined by the soap product used. Selecting a biodegradable, concentrated soap delivered via refillable reservoir rather than disposable cartridges would significantly reduce ongoing waste.

At the **end of life**, the product should be disassembled and its materials separated for appropriate disposal or recycling as outlined in Section 4.2.

4.2 Give Instructions on how to recycle.

When the Soapomorph prototype reaches the end of its usable life, it should be disposed of responsibly by following the steps below:

1. **Empty and clean** the soap reservoir before beginning disassembly to prevent contamination of recyclable materials
2. **Disassemble the unit** by separating the outer casing form the internal frame and carefully disconnect all electrical components
3. **Stainless steel components** should be taken to a scrap metal facility or household waste recycling centre (HWRC) with a metal collection point

4. **PLA** parts are plant-derived and technically compostable but require industrial composting conditions to break down properly. They should be taken to a specialist facility or HWRC that accept bioplastics
5. **PETG** parts carry recycling code 1 in some forms, though collection availability varies. PETG should be taken to a specialist plastics recycler or HWRC that accepts it
6. **ABS and polycarbonate** parts are both marked with recycling code 7 and are not accepted in standard kerbside recycling. These msut be taken to a specialist plastic recycler or an HWRC that accepts hard plastics
7. **Electrical components** must be disposed of as Waste Electrical and Electronic Equipment (WEEE). They must not be placed in general waste. Drop them off at a designated WEEE collection point
8. **Any remaining** non-recyclable components should be disposed of in general waste as a last resort

5.0 Materials.

5.1 Structure

The load-bearing structure of the electronic soap dispenser, including the internal support frame and mounting stand, shall be manufactured from stainless steel. Stainless steel is selected due to its high strength-to-weight ratio, corrosion resistance, and durability in wet and humid environments such as bathrooms and kitchens. As the dispenser is intended for frequent use and exposure to water, soap residue, and cleaning chemicals, the material must resist rust, degradation, and structural fatigue over time.

The structural frame must be rigid enough to support the moving dispenser assembly without flexing, as any structural instability could interfere with the alignment of the sliding mechanism. Mounting points for mechanical rails, hinges, and motor brackets must be precision-cut or machined to ensure consistent movement and long-term reliability.

The cosmetic outer frame (external casing) shall be manufactured from durable thermoplastic material produced via 3D printing during the prototyping phase. Suitable materials include PLA for early prototypes and ABS or PETG for functional testing due to their improved impact resistance and thermal stability. For final production, injection-moulded ABS or polycarbonate is recommended for improved durability, finish quality, and manufacturing efficiency.

The outer casing must provide both aesthetic appeal and protective housing for internal electrical and mechanical components. It shall be designed to prevent water ingress, protect users from moving parts, and allow access panels for maintenance and refill of soap reservoirs.

5.2 Mechanical Components

The moving dispenser mechanism, which enables the soap dispensing unit to travel back and forth, shall operate using a guided rail system. Linear rails or slider tracks should be manufactured from stainless steel or anodised aluminium to ensure smooth motion and corrosion resistance. Low-friction polymer bushings or linear bearings may be incorporated to reduce wear and noise during operation.

Hinge mechanisms, where applicable, shall also be stainless steel to maintain structural integrity under repetitive motion cycles. All moving joints must be designed to withstand repeated actuation without loosening or excessive mechanical play.

The drive system shall consist of either a servo motor or a DC motor coupled with a lead screw, belt drive, or rack-and-pinion mechanism depending on torque and precision requirements. A lead screw system is recommended for controlled linear motion and positional accuracy. The mechanism must ensure smooth, controlled movement without sudden jerks, as abrupt motion could reduce product lifespan or affect user experience.

Fasteners used throughout the mechanical assembly shall be stainless steel to prevent corrosion. Thread-locking compounds may be applied where necessary to prevent loosening due to vibration.

5.3 Electrical Components

The motor system shall consist of either a high-torque servo motor for precise positional control or a DC motor with encoder feedback for closed-loop motion control. The selected motor must provide sufficient torque to move the dispenser assembly under load while maintaining low power consumption.

All wiring shall use insulated copper cables rated for low-voltage DC systems. Cables must be securely routed within the casing using clips or channels to prevent interference with moving components.

The control system shall be managed by a microcontroller (such as an Arduino-compatible or similar embedded platform) responsible for motor control, sensor input processing, and timing functions. The microcontroller must be housed in a protected internal compartment to prevent moisture exposure.

Additional electrical components may include proximity sensors (infrared or ultrasonic) to detect hand placement, limit switches to define travel boundaries, and voltage regulation modules to ensure stable power delivery.

All electrical materials must comply with relevant safety standards and be rated for operation in humid environments.

6.0 Risk Assessment – Full Project Scope.

Risk Assessment Record	Project: Soapomorph				*To be assessed yearly*
Risk Assessed By:	Rachel Kirkham				
Name:	Group A	**Position:**	Student	**Location:**	Mellor Building.
Date:	22/02/2026	**Level:**	6	**Department:**	Department of Computing

Risk Matrix – Use this to determine level of risk for each hazard i.e. 'How bad & How likely'					
Severity of Harm	**(1) Rare** (e.g. once every few years)	**(2) Unlikely** (e.g. Once per year)	**(3) Possible** (e.g. Monthly)	**(4) Likely** (e.g. Weekly)	**(5) Very Likely** (e.g. daily or frequent)
(1) Negligible (e.g. Slight Bruising)	Low (1)	Low (2)	Low (3)	Medium (4)	Medium (5)
(2) Minor (e.g. Small cuts, deep bruising)	Low (2)	Low (4)	Medium (6)	Medium (8)	Medium (10)
(3) Moderate (e.g. Deep cuts, Torn muscles)	Low (3)	Medium (6)	Medium (9)	High (12)	High (15)
(4) Severe (e.g. Fracture, Loss of consciousness)	Medium (4)	Medium (8)	High (12)	High (16)	Extremely High (20)
(5) Very Severe	Medium (5)	High (10)	High (15)	Extremely High (20)	Extremely High (25)

(e.g. Death, Permanent disability/disease)								
Legend:	Low (1-4)		Medium (4-10)		High (11-15)		Extreme (16-25)	

Hazard List – List below.

Hazards identified	Risk	Potential injuries/ damage	Likelihood (1-5)	Severity (1-5)	Risk Rating (Likelihood x Severity)	Who is at risk	Additional measures/ Precautions made?	Who is responsible for action?
Electrical or battery fault	• Chemical Leak/ burns • Fire/ overheating • Electric shock • Fire	• Chemical burns • Laboratory damage • Electric shock • Shock • Burns	**2**	**4**	**8 (Medium)**	Team members	• All electrical equipment will have been PAT tested. • Pre-use inspection. • Switch on main power.	Rachel Kirkham Chris Mead Hannah Arnold Max McGill Balraj Bhuller
Animatronic Malfunction	• Unexpected or uncontrolled movement • Minor injury	• Bruising	**2**	**1**	**2 (Low)**	Team members	• Switch on main power. • Plenty of space to stand away from project.	Rachel Kirkham Chris Mead Hannah Arnold Max McGill Balraj Bhuller
Obstacles / Cluttered Area	• Trip/slip hazard • Collison • Blockage of exits	• Slip/ Trip • Injury to back • Sprain/ break • Head injury • Smoke inhalation • Blocked in fire • Minor injuries from cluttered workspace (such as dropping onto feet)	**2**	**4**	**8 (Medium)**	Team members	• There will be plenty of manoeuvrable space. • Exits and walkways kept clear. • Storage locations for belongings. • Workplace kept tidy.	Rachel Kirkham Chris Mead Hannah Arnold Max McGill Balraj Bhuller
Objects falling	• Heavy items falling • Tools falling.	• Breakages • Bruising • Fracture	**1**	**4**	**4 (Low)**	Team members	• Tidy workspace. No cluttered surfaces. • Storage area for belongings.	Rachel Kirkham Chris Mead Hannah Arnold Max McGill Balraj Bhuller

							• Appropriate use of PPE - steel toe caps.	
Cutting, welding metal.	• Burns. • Sparks. • Foreign objects. • Contact with equipment. • Bright/piercing lights. • Snagged clothing. • Untidy workplace. • Falling objects. • Damaged equipment.	• Blindness. • Eye injuries. • Severe burns. • Small cuts. • Deep cuts. • Injured limbs. • Damage to feet. • Long term damage.	**2**	**5**	**10 (Medium)**	Team members	• Wear appropriate PPE (Gloves, Welding mask, Goggles) • Wear appropriate tight clothing - tight fitted. • Wear steel toe cap safety boots. • Keep workstation tidy and all exits/ walkways tidy. • All equipment checked before and after use for wear and tear. • All equipment checked for PAT testing.	Rachel Kirkham.
Soldering.	• Contact with equipment. • Fire. • Untidy workplace. • Falling objects. • Fumes. • Foreign objects/ Objects in eye. • Damaged equipment.	• Eye injuries. • Severe burns. • Lung disease/ injury. • Electrocution. • Shock.	**2**	**5**	**10 (Medium)**	Team members	• Wearing of appropriate PPE (goggles and steel toe caps). • Wearing appropriate tight fitted clothing. • Stand up during soldering process to limit risk of solder dripping onto legs. • All equipment checked for wear and tear	Rachel Kirkham Chris Mead Hannah Arnold Max McGill Balraj Bhuller

							before and after use. • All equipment checked for PAT testing.	
Silicone moulding	• Toxic fumes. • Chemical exposure. • Poor ventilation. • Heat resistance failures. • Incomplete curing. • Incorrect manual handling. • Sharp tools and equipment. • Spills. • Cluttered work area.	• Skin irritation. • Chemical burns. • Allergic reactions. • Respiratory irritation. • Headaches. • Dizziness. • Nausea. • Skin blistering. • Minor to Severe burns. • Muscle or back strain. • Cuts/punctures. • Slips and falls. • Falling objects.	**2**	**4**	**8** **(Medium)**	Team members	• Wear appropriate PPE, gloves, goggles, lab coat, steel toe cap boots etc…. • Keep a well-ventilated area. • Heat sources supervised. • Keep workspace tidy and clean any spillages. • Wear gloves for sharp tool exposure. • Follow COSHH manufacturer data sheet. • Wash hands before and after (don't touch face during process).	Rachel Kirkham Chris Mead Hannah Arnold Max McGill Balraj Bhuller
Laser cutting	• Fire. • Poor ventilation. • Exposure to laser lighting. • Moving parts/mechanical injury/malfunctions. • Damages to equipment. • Damages to wiring or power source.	• Smoke/ toxic fume inhalation. • Eye strain to permanent damage/blindness. • Cuts/punctures. • Electrical shock. • Bruising. • Minor to major muscle injury. • Respiratory irritation/long term damage.	**1**	**4**	**4** **(Low)**	Team members	• Wear appropriate PPE, glasses if needed. • Make sure ventilation on machine works. • Check machine for PAT testing. • Check machine for external and internal damages.	Rachel Kirkham Chris Mead Hannah Arnold Max McGill Balraj Bhuller

	• Sharp edges/broken material handling.	• Headaches. • Dizziness. • Nausea.					• Check wiring to power socket. • Make sure emergency button is easily nearby. • Do not get material out of machine for 20 seconds to 1 minute to allow cooling and ventilation.	
3D printing	• Plastic fumes. • Fire. • Poor ventilation. • Heat buildup. • Malfunctions. • Exposed/damaged wiring. • Damages to machine present.	• Respiratory irritation/long term damage. • Headaches. • Dizziness. • Nausea. • Electrical shock. • Smoke/ toxic fume inhalation.	**1**	**4**	**4 (Low)**	Team members	• Check machine ventilation. • Check machine has been PAT tested. • Check internal and external parts of machine for signs of damage before and after use. • Check wiring and main power connection for signs of damage.	Rachel Kirkham Chris Mead Hannah Arnold Max McGill Balraj Bhuller
Fusion/blender	• Screen strain. • Incorrect posture long term.	• Eye strain. • Back / muscle strain.	**2**	**2**	**4 (Low)**	Team members	• Take consistent breaks throughout.	Rachel Kirkham Chris Mead Hannah Arnold Max McGill Balraj Bhuller
Assessor Signature:	R.Kirkham							
Date:	22/02/2026							

References

BBC Bitesize (2026) *Business stakeholders – revision 1*. Available at: https://www.bbc.co.uk/bitesize/guides/z6scbdm/revision/1 [Accessed: 24 February 2026]